"Closed Loop Control of Separately Excited DC Motor"

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Abstract:

In this project the mathematical model for closed loop control of separately excited dc motor is designed and tested through the MATLAB/Simulink software. In general, the high-performance motor drive is nothing but a motor drive in which a drive system should have good load regulating response and dynamic speed command tracking. Therefore, in acceleration and deceleration the dc motor provides excellent control in speed.
Keywords: Closed loop control, separately excited DC motor and DC motor.

INTRODUCTION

The variable speed, reliability and high performance are three main characteristics of an electric drive system due to which it can be easily controlled. The field of motor is connected directly to the power supply for the speed control. At the same time, it is necessary for torque and speed control. For low horsepower dc drives are low cost. In addition to this for overhauling loads dc regenerative drives are used.

By field control method and armature control method wide range of speed control both above and below the rated speeds are achieved. Therefore, dc motors are used in fine speed applications such as in paper mills and in rolling mills. In general, on the basis of dc motor excitation the dc motor is classified into two types. They are separately excited and self-excited dc motor. In this project we used separately excited dc motor. Hence its field winding and armature are excited from two different sources. The fundamental of electric drives, power electronic circuits, devices and application are explained in detail.

CONSTRUCTION AND WORKING PRINCIPLE OF DC MOTOR

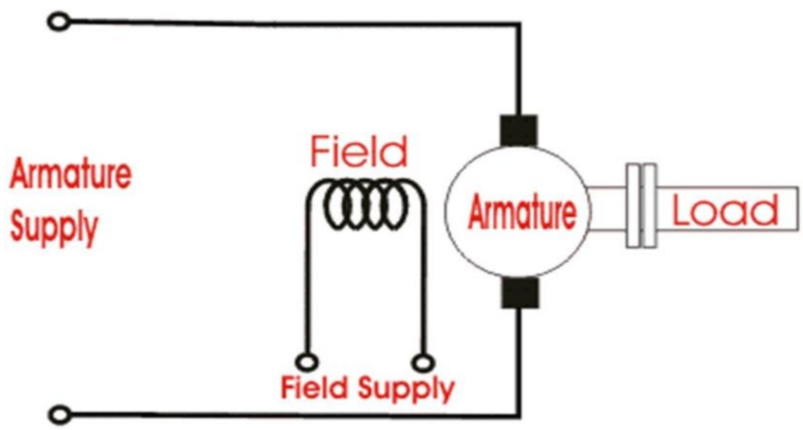

Figure {1} DC motor

In the figure {1} with separate supply the separately excited dc motor has field and armature winding. To excite the field, flux the field winding of the dc motor is used. For mechanical work current in armature circuit is supplied to the rotor through brush and commutator. The field flux and armature current interaction produces rotor torque. Okbuka and raju singh has explained about the performance characteristics of controlled separately excited dc motor and stability analysis of separately excited dc motor respectively [4-5]. The working principle of separately excited dc motor is listed in points below

Point 1: The field current {if} excites the separately excited dc motor
Point 2: In the circuit armature current {ia} flows
Point 3: To balance load torque at particular speed. The motor develops a back emf and torque.
Point 4: Any change in armature current has no effect on the field current {if independent of ia}.
Point 5: The field current {if} is much less than the armature current

MATHEMATICAL EQUATIONS OF SEPERATELY EXCITED DC MOTOR

Armature equation,

$$Equation.1$$

$$\frac{di_a}{dt} = -\frac{R_a}{L_a} i_a - \frac{K_{bt}}{L_a} \omega_a + \frac{V}{L_a}$$

Torque equation,

$$Equation.2$$

$$\frac{d\omega_a}{dt} = -\frac{B}{J_m} \omega_a + \frac{K_{bt}}{J_m} i_a - \frac{T_L}{J_m}$$

Where,

V : Voltage applied to armature in Volts
i_a : Armature current in Ampere
R_a : Armature resistance in Ohm
L_a : Inductance of the armature in Henry
T_L : Load torque in N-m
ω_a : Mechanical speed of the motor in rad/sec
B : Damping coefficient
J_m : Moment of inertia of the motor
K_{bt} : Machine coefficient

SIMULATION MODEL OF DC MOTOR

Armature resistance {R_a} =1.8 ohms
Armature inductance {L_a} =0.0063H
Moment of inertia { J_m } =0.08735 kg/m^2
Machine coefficient { K_{bt} } =4.212 v/rad
Rated terminal voltage {V_a} =220 volt
Damping coefficient {Bm} = 0.023 nm/rad/s
Rated armature current {I_a} = 13.63 Amps
Load torque {T_L} =19.07 N-m

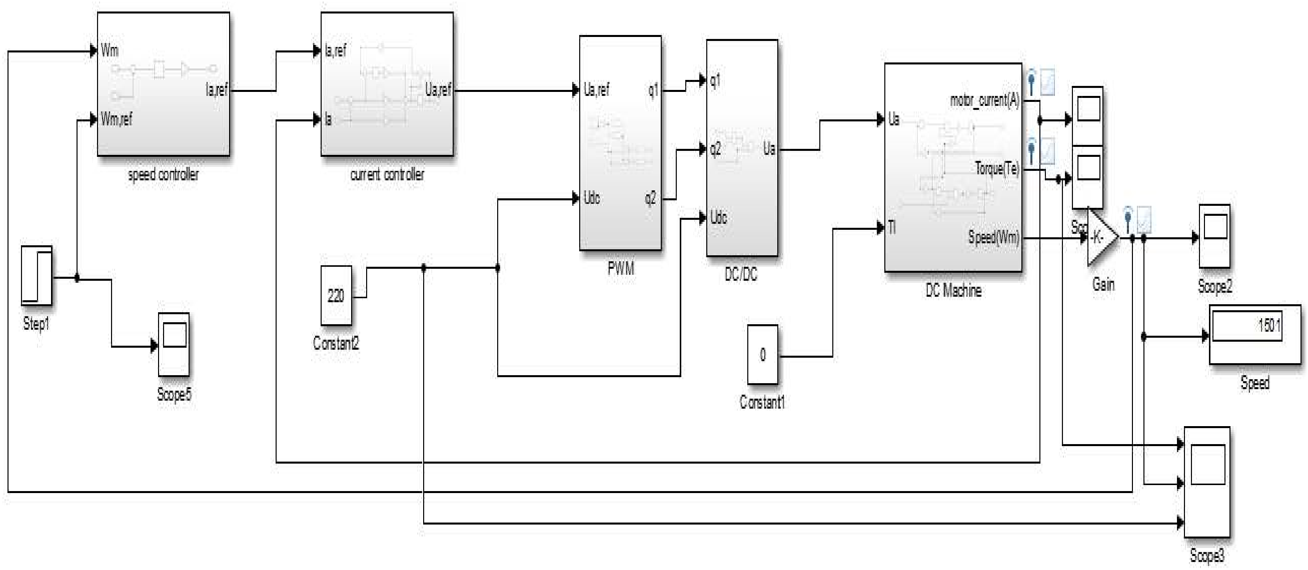

Figure {2} Simulation model of closed loop separately excited dc motor

The output speed of the separately excited DC motor is compared with given reference speed. From the comparison, the speed error and change in speed error are calculated and given as input to the controllers. Output of the controller is given to the current controller as reference current. The current controller performs the comparison between reference current and actual armature current of DC motor. Based on the comparison, it generates the gate signal for chopper drive which is turn controls the input voltage given to the motor.

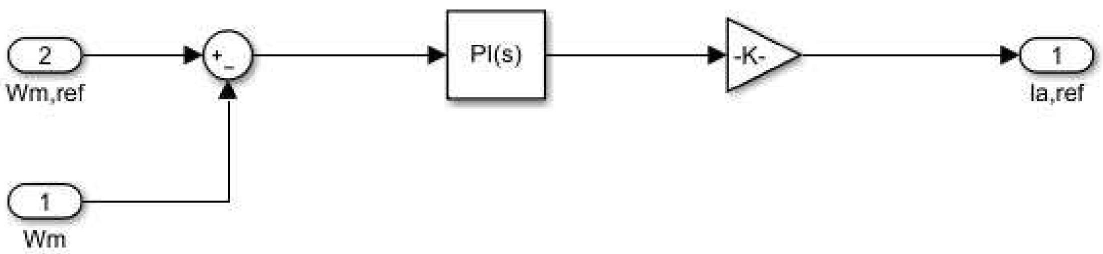

Figure {3} speed controller subsystem

The speed controller is present in the outer loop. An increase in the reference speed ω_m produce a positive error $\Delta\ \omega_m$. Speed error is processed through the speed controller and further given to next subsystem .

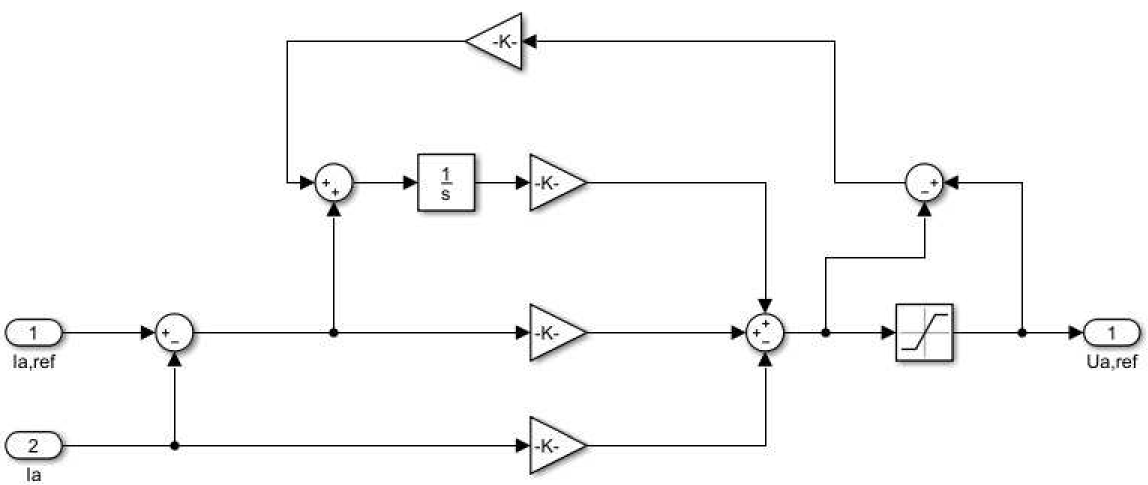

Figure {4} Current controller subsystem

Current controller is present in the inner loop. It is provided to limit the converter and motor torque below a safe limit.

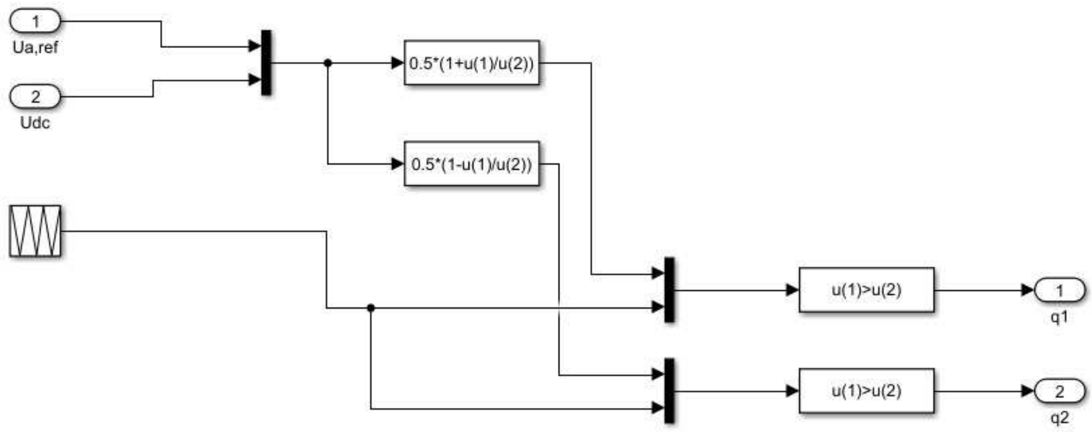

Figure {5} PWM subsystem

The PWM sub system is present to give the PWM pulses to the DC-DC converter. The converter requires the pulses for switching of the devices present in it.

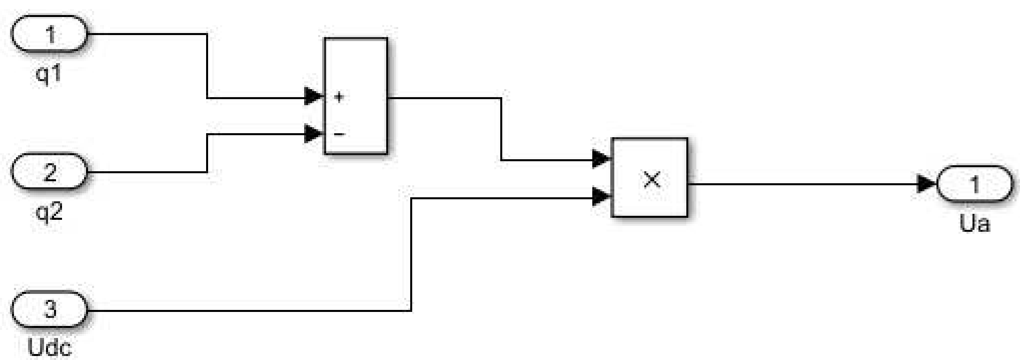

Figure {6} DC/DC subsystem

This figure represents the operation of the DC-DC converter that converts DC voltage from one level to another level. It receives the pulses required for the conversion through the PWM subsystem.

Simulation results:

Case 1: Step disturbance in reference speed from 50% rated speed to rated speed.(Performing this at no load condition and at rated voltage.)

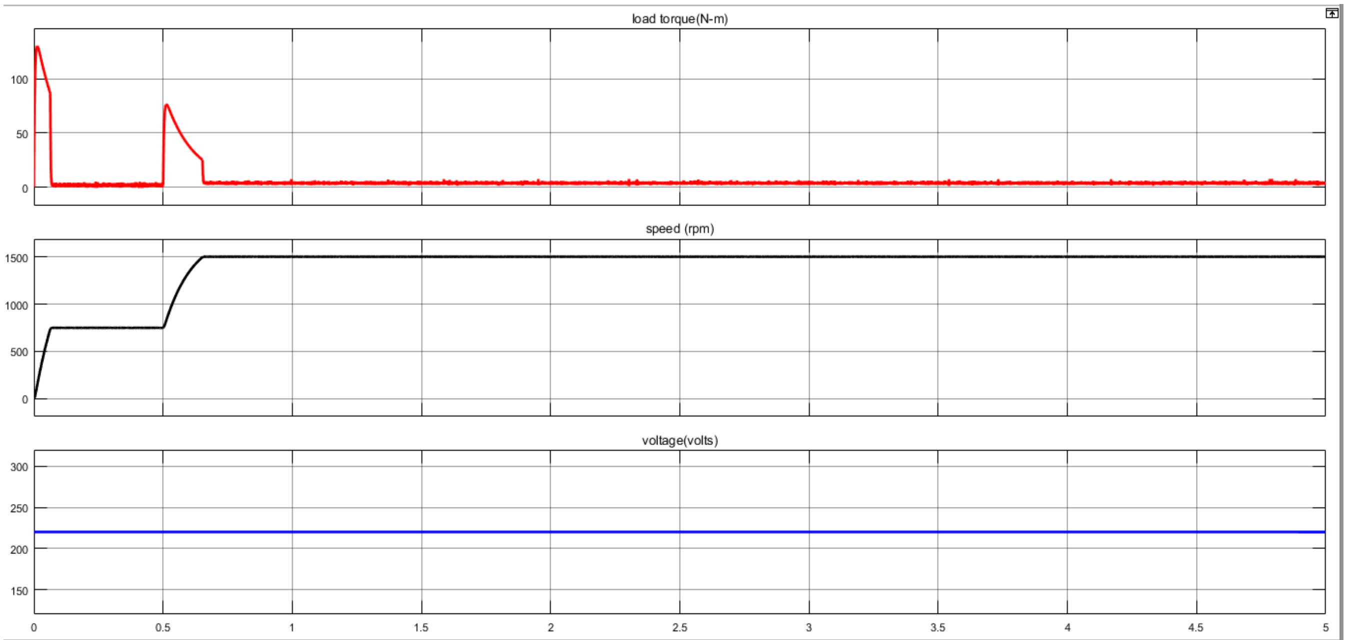

Figure {7} graphical representation of the response given by the Separately Excited DC Motor at Step disturbance in reference speed from 50% rated speed to rated speed.

The figure shows the graphical representation of the response given by the Separately Excited DC Motor at Step disturbance in reference speed from 50% rated speed to rated speed at no load condition and at rated voltage. The first curve in the red represent the variation of the load torque, the second curve in the black represents the variation of the speed, the third curve in the blue represents the variation of the voltage.

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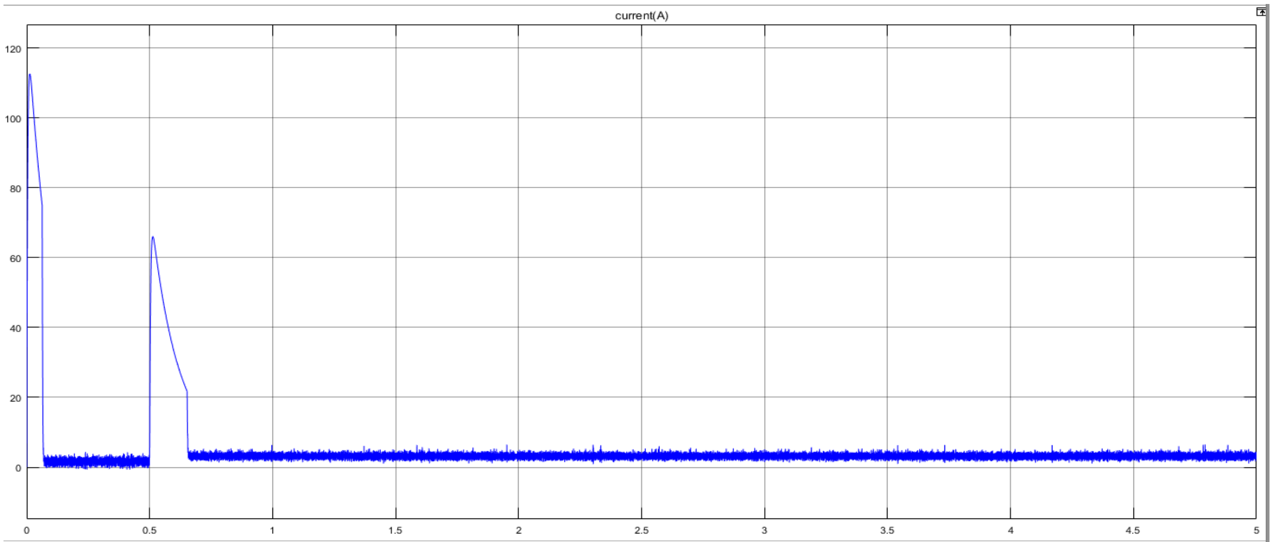

Figure {8} graphical representation of the response given by the Separately Excited DC Motor's current at Step disturbance in reference speed from 50% rated speed to rated speed

The curve in the figure above represents the variation in the current when there is Step disturbance in reference speed from 50% rated speed to rated speed

<u>Case 2</u>: Step disturbance in input voltage from 80% of the rated voltage to100%.(Performing this at no load condition and at rated speed.)

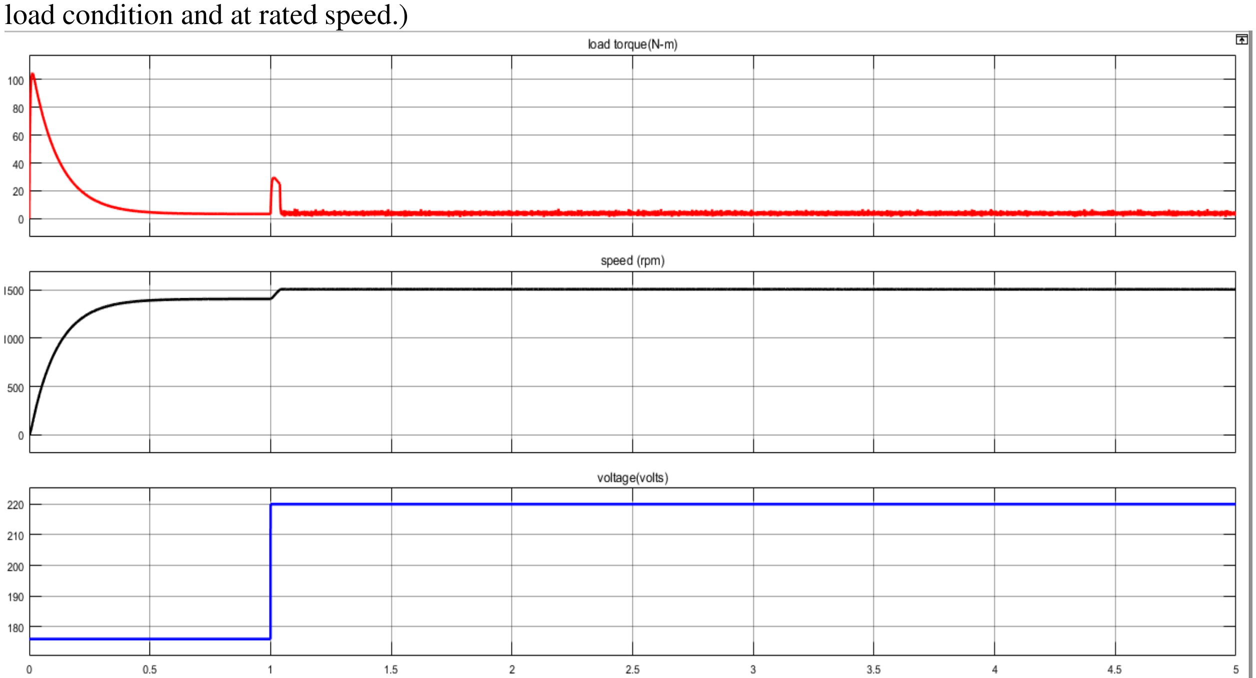

Figure {9} graphical representation of the response given by the Separately Excited DC Motor at Step disturbance in input voltage from 80% of the rated voltage to100%.

The figure shows the graphical representation of the response given by the Separately Excited DC Motor at Step disturbance in input voltage from 80% of the rated voltage to100% at no load condition and at rated voltage. The first curve in the red represent the variation of the load torque.

The second curve in the black represents the variation of the speed, The third curve in the blue represents the variation of the voltage.

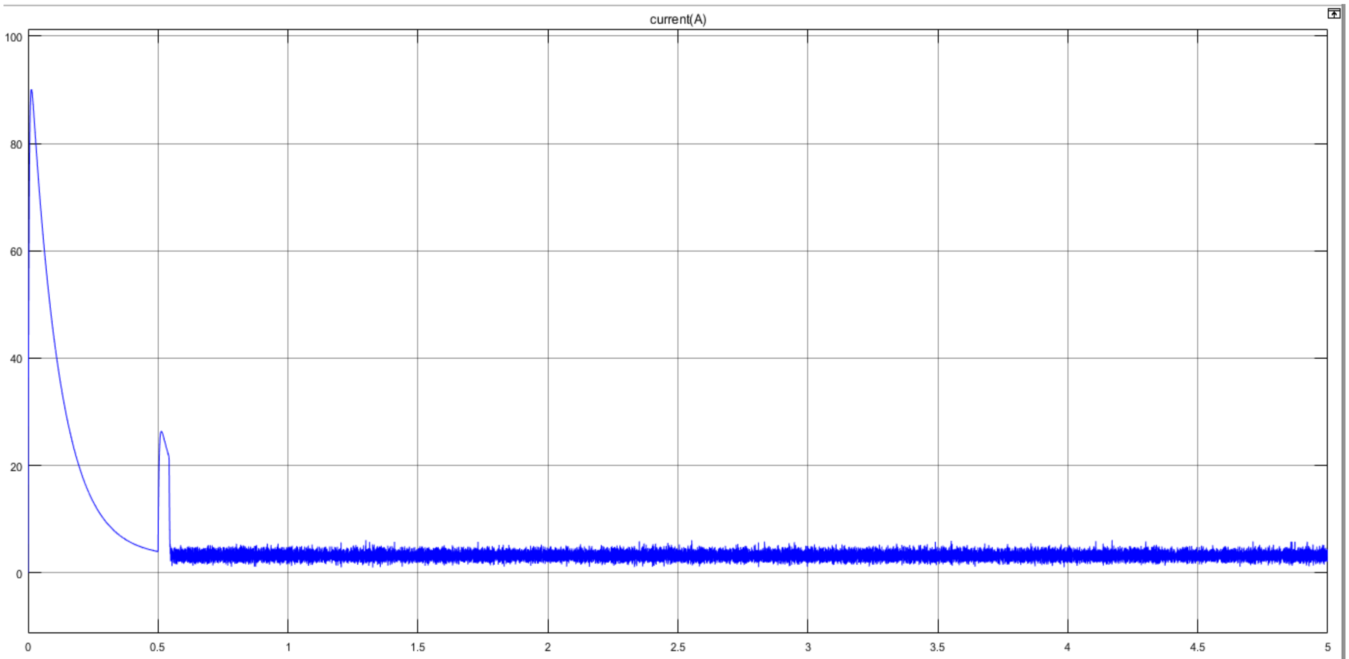

Figure {10} graphical representation of the response given by the Separately Excited DC Motor's current at Step disturbance in input voltage from 80% of the rated voltage to100%.

The curve in the figure above represents the variation in the current when there is Step disturbance input voltage from 80% of the rated voltage to100%.

Case 3: Step disturbance in load from no load to 50% of full load .(Performing this at rated voltage and at rated speed condition.)

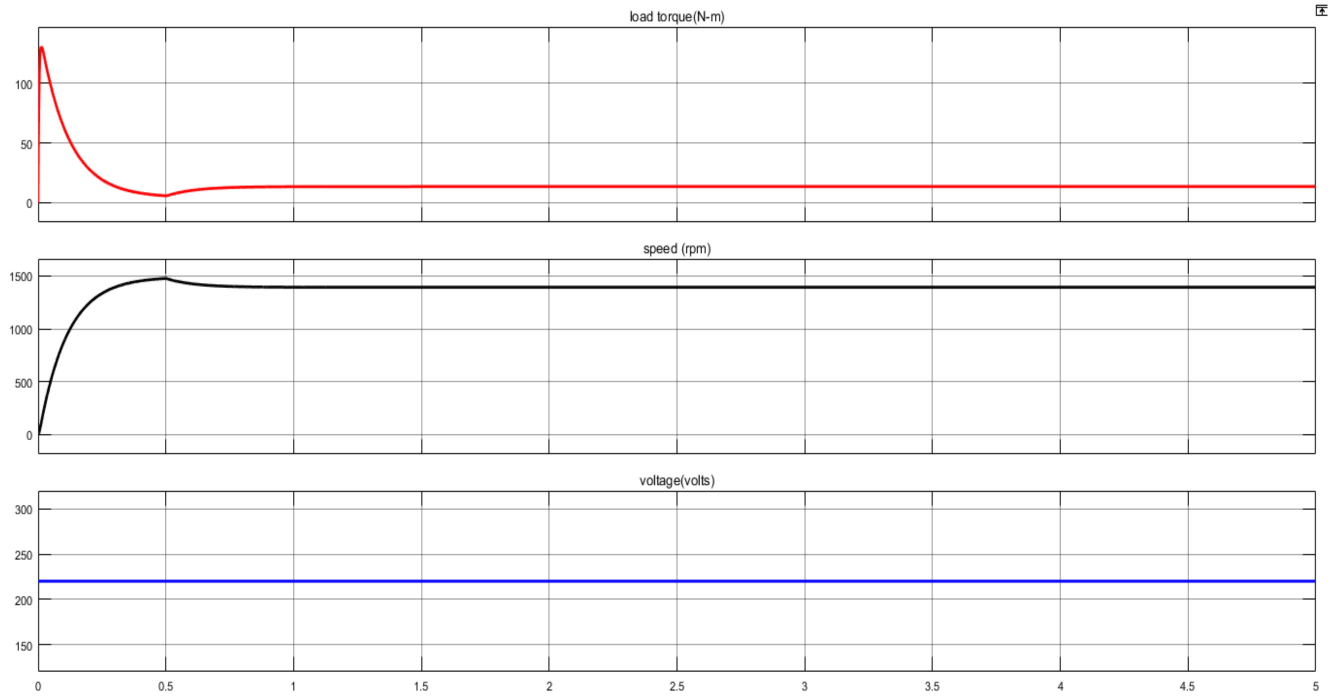

Figure {11} graphical representation of the response given by the Separately Excited DC Motor at Step disturbance in in load from no load to 50% of full load.

The figure shows the graphical representation of the response given by the Separately Excited DC Motor at Step disturbance in load from no load to 50% of full load at rated voltage and at rated speed condition. The first curve in the red represent the variation of the load torque, The second curve in the black represents the variation of the speed, The third curve in the blue represents the variation of the voltage.

Tasks:

Task1: The mathematical model is developed using the set of integral differential equations describing the functions of the separately excited DC motor and it is shown above.

Task 2: Different types of speed controllers available for the speed control are

fuzzy logic controller, fuzzy PI controller, PSO based tuning of Fuzzy PI controller and the new hybrid algorithm of PSODE based of Fuzzy PI can control the speed

The inputs to the **FLC** are speed error and change in speed. Speed error is defined as the difference between the actual speed and reference speed of the motor. Seven membership functions are created for each input and output. The FLC has been built with two inputs and one output. The membership functions are Negative Small, Negative Medium, Negative Big, Zero, Positive Small, Positive Medium, and Positive Big. Based on the values error and change in error of speed, the output of the FLC is in terms of current. This is the reference value for the current controller. The difference between the reference current from the fuzzy controller and actual armature current is given as an input to the current controller. Based on the current limit, the current controller generates the gate current to the chopper drive. The armature voltage of the separately excited DC motor is varied when variation occurs in the gate pulse of the chopper. Thus, the speed of the motor is controlled.

In **fuzzy PI** controller, the gain value is added with speed error and change in speed error. Gain value is used as P and I value of PI controller.

In **PSO tuned the fuzzy PI controller**, the range of membership functions are tuned to achieve better speed control than fuzzy and fuzzy PI controllers. Each membership function of the inputs and output is considered as a particle. So, the number of particles for the controller is 21. Mean Square Error (MSE) is considered as the fitness function for the PSO algorithm. The minimum of MSE is obtained by using PSO algorithm. The range of membership functions are tuned by PSO, and based on the result of PSO tuning, the range of input and output membership functions are changed toobtain minimum value for the fitness function

The other Different methods for speed control of DC motor:

- Traditionally armature voltage using Rheostatic method for low power dc motors.
- Use of conventional PID controllers.
- Neural Network Controllers.
- Constant power motor field weakening controller based on load-adaptive multi-input multi-output linearization technique (for high speed regimes).
- Single phase uniform PWM ac-dc buck-boost converter with only one switching device used for armature voltage control.
- Using NARMA-L2 (Non-linear Auto-regressive Moving Average) controller for the constant torque region.

Task 3:

The combination of proportional and integral terms is important to increase the speed of the response and also to eliminate the steady state error. The transfer function of PI controller has the form KP +KI⁄S,where KP is proportional gain and KI is an Integral gain. The proportional term (sometimes called gain) makes a change to the output that is proportional to the current error value. The proportional response can be adjusted by multiplying the error by a constant Kp, called the proportional gain.

Proportional-Integral (PI) controller is often to control the motor speed of the DC motor. The major features of the PI controller are its simplicity and ability to maintain a zero steady-state error to a step change in reference. So we have chosen the PI controller.

Task 4:

The nature of the load being used here is **Torque proportional to the Speed.**

The magnitude of the load being used is 19.07 N-m.

Because,

$P = \omega . T$ …………………(1)

$P = \frac{2\pi n T}{60}$ ………………….(2)

Given :

P=3KW

N=1500 RPM

So we can calculate T from the equation (2)

Therefore,

$3000 = 2\pi(1500)T/60$

$\frac{3000X60}{2\pi X1500} = T$

$\boldsymbol{T = 19.07}$ **N-m**

CONCLUSION

The closed control loop for speed regulation of separately excited DC motor for the disturbances mentioned in the problem has been done successfully.
The nature of the torque, current, voltage and speed for the given disturbance is obtained

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